A Technical Introduction to Bias variance tradeoff

$$\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Variance} + \sigma_\varepsilon^2$$

Section 1

In human learning While widely discussed in the context of machine learning, the bias–variance dilemma has been examined in the context of human cognition, most notably by Gerd Gigerenzer and co-workers in the context of learned heuristics. They have argued (see references below) that the human brain resolves the dilemma in the case of the typically sparse, poorly-characterized training-sets provided by experience by adopting high-bias/low variance heuristics. This reflects the fact that a zero-bias approach has poor generalizability to new situations, and also unreasonably presumes precise knowledge of the true state of the world. The resulting heuristics are relatively simple, but produce better inferences in a wider variety of situations. Geman et al. argue that the bias–variance dilemma implies that abilities such as generic object recognition cannot be learned from scratch, but require a certain degree of "hard wiring" that is later tuned by experience. This is because model-free approaches to inference require impractically large training sets if they are to avoid high variance.

Section 2

The expectation ranges over different choices of the training set D = { (x 1 , y 1) … , (x n , y n) } {\displaystyle D=\{(x_{1},y_{1})\dots ,(x_{n},y_{n})\}} , all sampled from the same joint distribution P (x , y) {\displaystyle P(x,y)} which can for example be done via bootstrapping. The three terms represent:

Section 3

MSE (x) ■ E [(y – f ^ (x)) 2] = E [(f (x) + ε – f ^ (x)) 2] since y ■ f (x) + ε = E [(f (x) – f ^ (x)) 2] + 2 E [(f (x) – f ^ (x)) ε] + E [ε 2] {\displaystyle {\begin{aligned}{\text{MSE}}(x)&\triangleq \mathbb {E} {\Big [}{\big (}y-{\hat {f}}\!(x){\big)}^{2}{\Big]}\\&=\mathbb {E} {\Big [}{\big (}f(x)+\varepsilon -{\hat {f}}\!(x){\big)}^{2}{\Big]}&&{\text{since }}y\triangleq f(x)+\varepsilon \\&=\mathbb {E} {\Big [}{\big (}f(x)-{\hat {f}}\!(x){\big)}^{2}{\Big]}\,+\,2\ \mathbb {E} {\Big [}{\big (}f(x)-{\hat {f}}\!(x){\big)}\varepsilon {\Big]}\,+\,\mathbb {E} [\varepsilon ^{2}]\end{aligned}}}

Section 4

E D , ε [(y – f ^ (x ; D)) 2] = (Bias D ■ [f ^ (x ; D)]) 2 + Var D ■ [f ^ (x ; D)] + σ 2 {\displaystyle \mathbb {E} _{D,\varepsilon }{\Big [}{\big (}y-{\hat {f}}\!(x;D){\big)}^{2}{\Big]}={\Big (}\operatorname {Bias} _{D}{\big [}{\hat {f}}\!(x;D){\big]}{\Big)}^{2}+\operatorname {Var} _{D}{\big [}{\hat {f}}\!(x;D){\big]}+\sigma ^{2}}

Section 5

E ■ [(f (x) – E ■ [f ^ (x)]) (E ■ [f ^ (x)] – f ^ (x))] = E ■ [f (x) E ■ [f ^ (x)] – f (x) f ^ (x) – E ■ [f ^ (x)] 2 + E ■ [f ^ (x)] f ^ (x)] = f (x) E ■ [f ^ (x)] – f (x) E ■ [f ^ (x)] – E ■ [f ^ (x)] 2 + E ■ [f ^ (x)] 2 = 0 {\displaystyle {\begin{aligned}&{\color {PineGreen}\operatorname {\mathbb {E} } \left[\left(f(x)-\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]\right)\left(\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]-{\hat {f}}\!(x)\right)\right]}\\&=\operatorname {\mathbb {E} } \left[f(x)\,\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]\,-\,f(x){\hat {f}}\!(x)\,-\,\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]^{2}+\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]\,{\hat {f}}\!(x)\right]\\&=f(x)\,\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]\,-\,f(x)\,\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]\,-\,\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]^{2}\,+\,\operatorname {\mathbb {E} } [{\hat {f}}\!(x)]^{2}\\&=0\end{aligned}}}

Section 6

In regression The bias–variance decomposition forms the conceptual basis for regression regularization methods such as LASSO and ridge regression. Regularization methods introduce bias into the regression solution that can reduce variance considerably relative to the ordinary least squares (OLS) solution. Although the OLS solution provides non-biased regression estimates, the lower variance solutions produced by regularization techniques provide superior MSE performance.

Section 7

MSE = E x ■ [MSE (x)] = E x ■ { Bias D [f ^ (x ; D)] 2 + Var D ■ [f ^ (x ; D)] } + σ 2 . {\displaystyle {\text{MSE}}=\operatorname {\mathbb {E} } _{x}\left[{\text{MSE}}(x)\right]=\operatorname {\mathbb {E} } _{x}\left\{\operatorname {Bias} _{D}\!\left[{\hat {f}}\!(x;D)\right]^{2}+\operatorname {Var} _{D}\left[{\hat {f}}\!(x;D)\right]\right\}+\sigma ^{2}.}

Section 8

In Monte Carlo methods While in traditional Monte Carlo methods the bias is typically zero, modern approaches, such as Markov chain Monte Carlo are only asymptotically unbiased, at best. Convergence diagnostics can be used to control bias via burn-in removal, but due to a limited computational budget, a bias–variance trade-off arises, leading to a wide range of approaches, in which a controlled bias is accepted, if this allows to dramatically reduce the variance, and hence the overall estimation error.

Section 9

Var D ■ [f ^ (x ; D)] ■ E D [(E D [f ^ (x ; D)] − f ^ (x ; D)) 2] {\displaystyle \operatorname {Var} _{D}{\big [}{\hat {f}}\!(x;D){\big]}\triangleq \mathbb {E} _{D}\left[\left(\mathbb {E} _{D}[{\hat {f}}\!(x;D)]-{\hat {f}}\!(x;D)\right)^{2}\right]}

Section 10

E ■ [(y − f ^ (x)) 2 ■ X = x] = (f (x) − 1 k ∑ i = 1 k f (N i (x))) 2 + σ 2 k + σ 2 {\displaystyle \operatorname {\mathbb {E} } \left[\left(y-{\hat {f}}\!(x)\right)^{2}\mid X=x\right]=\left(f(x)-{\frac {1}{k}}\sum _{i=1}^{k}f(N_{i}(x))\right)^{2}+{\frac {\sigma ^{2}}{k}}+\sigma ^{2}}

Section 11

In reinforcement learning Even though the bias–variance decomposition does not directly apply in reinforcement learning, a similar tradeoff can also characterize generalization. When an agent has limited information on its environment, the suboptimality of an RL algorithm can be decomposed into the sum of two terms: a term related to an asymptotic bias and a term due to overfitting. The asymptotic bias is directly related to the learning algorithm (independently of the quantity of data) while the overfitting term comes from the fact that the amount of data is limited.